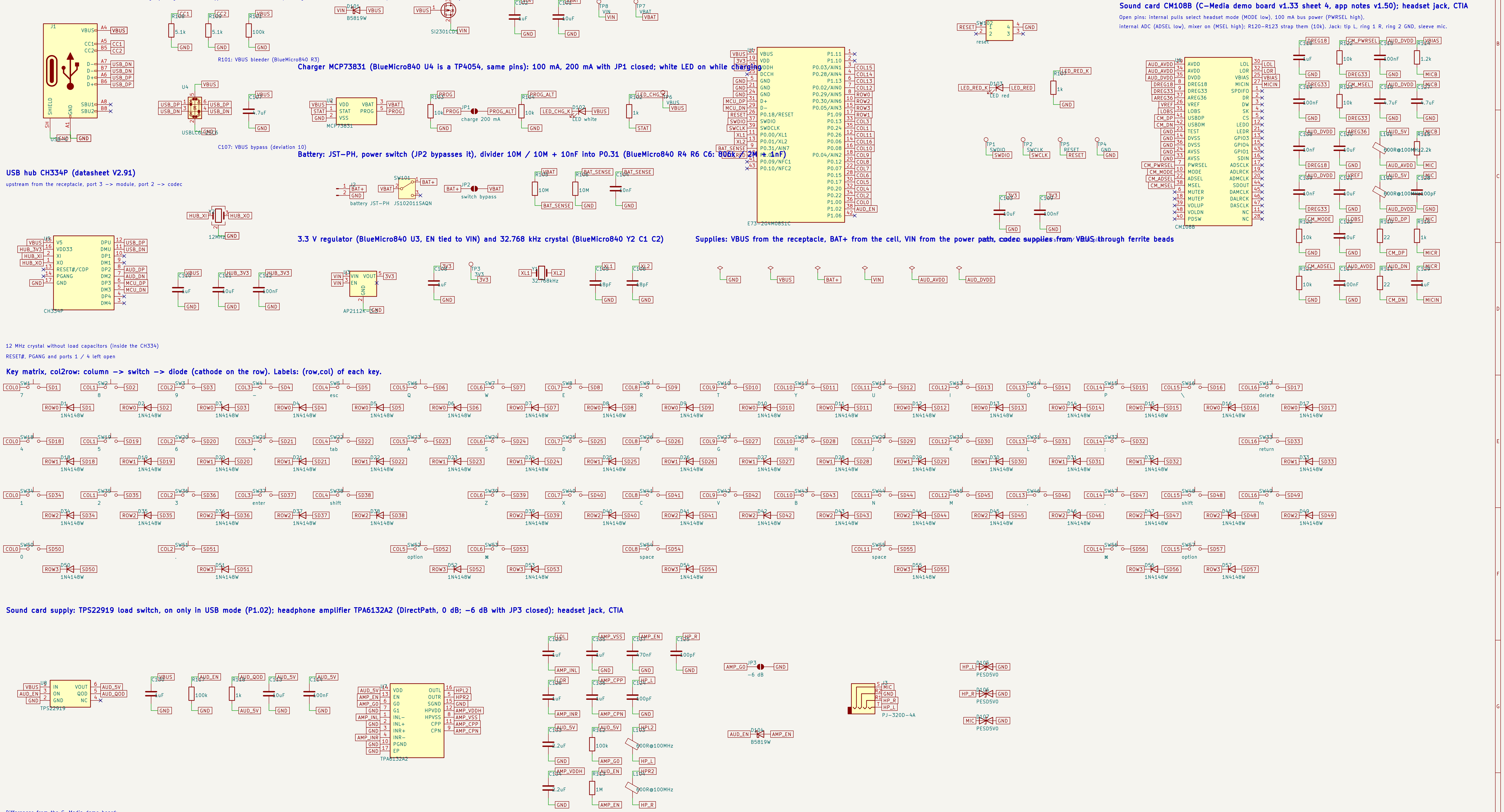
Lira: nRF52840 (Ebyte E73–2G4M08S1C), USB–C + Li–Po, 57 keys, built–in USB sound card

Power / USB / programming copied from the BlueMicro840 (jpconstantineau/NRF52–Board), a board built with this module, assembled by JLCPCB and sold. Differences from it:

1. Second AP2112K (switched 3.3 V for external devices, P0.12 enable) omitted: nothing external to power.
2. USBLC6–2SC6 ESD array added on D+ / D– / VBUS at the USB–C receptacle.
3. Power switch SW101 (C&K JS102011SAQN, 300 mA) in the battery + line, as on ZMK keyboards; JP2 bypasses it.
4. JP1 + R104: closing the solder jumper puts a second 10k on PROG: charge current 100 mA –> 200 mA.
5. Reset push button SW102 to GND (the BlueMicro840 has only a test point).
6. USB–C shield pads to GND (left open on the BlueMicro840).
7. LEDs: charging LED white, status LED red on P1.04 only (BlueMicro840: green charging, red P1.04, blue P1.10).
8. DCCH inductor pads (DNP on the BlueMicro840) not fitted at all; P1.10 is a free GPIO here, P1.02 switches the sound card.
9. Test points added on VBUS, VBAT and VIN (BlueMicro840: SWDIO, SWCLK, VCC, GND, RESET, P1.02).
10. C107 4.7 uF added on VBUS at the charger's VDD pin (the MCP73831 datasheet asks for 4.7 uF; Nordic's reference circuit has 4.7 uF on VBUS; the BlueMicro840 has none).
11. C108 1 uF at the regulator output and C109 100 nF at the module: the 10 uF C103 sits at the module's supply pins (VDD / VDDH), 25 mm away. C102 on VIN is 1 uF instead of 10 uF: together they keep the capacitance charged at USB attach near USB 2.0's 10 uF.
12. Charger MCP73831 instead of the TP4054, same pins and same PROG resistor (1000 V / R_PROG): it is stable with no battery, so the board also works when built without one (its STAT pin, and the white LED, stay off).
13. Battery divider 10M + 10M with 10 nF (BlueMicro840: 806k + 2M, 1 nF): the MCP73831 recognises a missing battery only if VBAT sees more than 7 MOhm (datasheet 5.2); 10 nF >= 2^12 x the SAADC's 2.5 pF keeps the reading within 1 LSB (Nordic, 'Measuring Lithium battery voltage with nRF52'), and RMK samples it every 30 s. Drain 0.2 uA.
14. Built–in USB sound card: the USB–C data lines go (through the ESD array) to a CH334P hub; hub port 3 goes to the module's D+ / D–, port 2 to a CM108B codec (C–Media demo board v1.33) with a 3.5 mm headset jack.



USB–C (sink: 5.1k on each CC) and ESD protection

R101: VBUS bleeder (BlueMicro840 R3)

C107: VBUS bypass (deviation 10)

Power path: USB through the Schottky, battery through the P–MOSFET (off while VBUS is present)

nRF52840 module, normal–voltage mode (VDD = VDDH = 3.3 V); reset button; status LED on P1.04 (bootloader LED); SWD pads

Sound card CM108B (C–Media demo board v1.33 sheet 4, app notes v1.50); headset jack, CTIA

Open pins: internal pulls select headset mode (MODE low), 100 mA bus power (PWRSEL high), internal ADC (ADSEL low), mixer on (MSEL high); R120–R123 strap them (10k). Jack: tip L, ring 1 R, ring 2 GND, sleeve mic.

Charger MCP73831 (BlueMicro840 U4 is a TP4054, same pins): 100 mA, 200 mA with JP1 closed; white LED on while charging

Battery: JST–PH, power switch (JP2 bypasses it), divider 10M / 10M + 10nF into P0.31 (BlueMicro840 R4 R6 C6: 806k / 2M + 1nF)

USB hub CH334P (datasheet V2.91)

upstream from the receptacle, port 3 –> module, port 2 –> codec

12 MHz crystal without load capacitors (inside the CH334)

RESET#, PGANG and ports 1 / 4 left open

3.3 V regulator (BlueMicro840 U3, EN tied to VIN) and 32.768 kHz crystal (BlueMicro840 Y2 C1 C2)

Supplies: VBUS from the receptacle, BAT+ from the cell, VIN from the power path, codec supplies from VBUS through ferrite beads

C103, C109 at the module's VDD / VDDH pins

Key matrix, col2row: column –> switch –> diode (cathode on the row). Labels: (row,col) of each key.

Sound card supply: TPS22919 load switch, on only in USB mode (P1.02); headphone amplifier TPA6132A2 (DirectPath, 0 dB; –6 dB with JP3 closed); headset jack, CTIA

Differences from the C–Media demo board:

1. Supply: VBUS through a TPS22919 load switch (soft start 1.3 ms, output discharge) to AUD_5V, ON = P1.02 with a 100k pull–down: the sound card is powered only in USB mode (firmware src/audio_gate.rs). A hard switch–on would dip VBUS enough to reset the hub (CH334 datasheet 6.1).
2. Headphones: a TPA6132A2 DirectPath amplifier after LOL / LOR (demo board: 100 uF + 39R straight to the jack): about 0 ohm output impedance, no coupling capacitors, 0 dB gain (–6 dB with JP3 closed, for very sensitive earphones). Inputs single–ended through 1 uF (8 Hz); EN about 0.25 s after the supply (1M + 470 nF from P1.02) and off at once through a Schottky. The switch's output discharge goes through 1k (TPS22919 9.2.2.2): the 5 V falls in about 3 ms, so the amplifier is off before the codec loses its supply (TPA6132A2 p.12, no pop).
3. Microphone: bias 1.2k + 2.2k (demo board 2.2k + 8.2k, for PC microphones) for phone headsets (Android: 1.8–2.9 V through 2.2k); 1k in series before MICIN against the step when a plug goes in.
4. The demo board's two 10 nF capacitors DVDD – DREG18 and DVDD – DREG33 ('close to IC', for ESD) are fitted.
5. The amplifier's SGND (pin 15) is grounded at the amplifier (its ground pad and a via into the top ground plane), not on a track of its own to the jack's ground pin as TI suggests (datasheet 10.3): with both layers poured the difference is a few millivolts.
6. The four mode jumpers are also strapped with 10k (MODE/ADSEL low, PWRSEL/MSEL high to DREG33) on top of the chip's internal pulls. D105–D107 are bidirectional ESD clamps (PESD5V0S1BA) on the jack's left, right and microphone contacts: a DirectPath output swings below GND, so the clamp must

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